MIDTERM RESEARCH PROPOSAL

Impact of Weather Information Availability on One-Hour-Ahead Bike-Sharing Demand Forecasting: A Cross-Model Study

Predictive Learning

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October 27, 2026

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Research Background and Motivation

> Weather is a recognized driver of bike-sharing use, but a **forecast** can only use weather that is already known.

Demand depends on weather

Cold, rain and high humidity are associated with fewer and shorter bikeshare trips (Gebhart & Noland, 2014). Weather and temporal factors recur across the bike-sharing literature (Eren & Uz, 2020).

Seoul hourly data is public

The Seoul Bike Sharing Demand data pairs hourly rental counts with weather and calendar fields, and has been used for weather-based demand prediction (Sathishkumar & Cho, 2020).

Evaluation must match use

Hourly forecasts are made at a forecast origin and applied to later months. Which weather is available, and how flexible the model is, should be judged out of time.

Scope: system-wide hourly rental counts only; no station, return-flow or inventory data, so no rebalancing claims.

Research Problem and Forecasting Scenario

> Which information is available at the **forecast origin**, and how complex should the model be?

Forecasting scenario

- **Target:** system-wide Rented Bike Count in the target hour (rentals per hour)
- **Forecast origin:** end of the previous hour; only earlier observations may be used
- **Protocol:** rolling one-step forecasts; model parameters stay frozen during testing

P1 Information availability

Is previous-hour observed weather useful? How much does target-hour actual weather change the picture?

P2 Model complexity

How do regularization and flexibility affect generalization from winter–spring training to later months?

P3 Attribution and errors

What do demand history, time, calendar and weather each add, and when do forecasts fail?

Related Work: Bike-Sharing Demand Forecasting

Study	Setting	What it contributes
Eren & Uz (2020)	Review of station-based bike-sharing research	Organizes factors behind demand: weather, built environment, public transport, socio-demographics, temporal and safety factors.
Sathishkumar & Cho (2020)	Seoul hourly rentals with weather data	Weather-based, rule-based model for Seoul bike demand prediction; this line of work uses the Seoul Bike dataset adopted here.
Xin (2026)	Seoul and Washington, DC; one-hour-ahead	Probabilistic forecasting with Transformer-based and other models; reports point error and 80% interval coverage.

> Prior work links weather to demand and forecasts Seoul hourly demand with learned models.

Sources: see References (slide 14). Descriptions summarize each work's stated scope.

Related Work: Weather Information and Predictive Models

Weather evidence

- Gebhart & Noland (2014), Washington, DC: cold temperature, rain and high humidity reduce both the likelihood and duration of bikeshare trips.
- Eren & Uz (2020): weather is one of several factor groups studied for demand.
- Sathishkumar & Cho (2020): weather variables used for Seoul demand prediction.

Predictive models: Xin (2026), Seoul, one hour ahead

Model	MAE (rentals/h)	80% coverage
N-BEATS	115.07	75.42%
Random Forest	115.86	89.26%
TFT	306.98	81.11%
Transformer	317.72	74.01%

Reported by Xin (2026), Table 2 (p. 6). Not our results and not directly comparable to ours.

> Lowest point error does not imply the best interval coverage.

Open point: the literature does not specify, field by field, which weather values exist at the forecast origin of a one-hour-ahead forecast.

Research Gap and Proposed Contribution

Insufficiently clarified question	What prior work leaves open	Our contribution
Information timing	Xin (2026) describes target-time covariates; availability of each field at the forecast origin is not fully specified (not an allegation of leakage).	Explicit E1 / E2 / E3 information settings under one protocol
Complexity control	Models from different families are compared; controlled complexity curves are not the focus.	Ridge α, KNN k and tree-depth curves with train vs. validation error (X2, X3)
Information value	Which input group matters, and where errors concentrate, is not isolated.	Group ablation and temporal error slices (X4, X5)

Shared with Xin (2026): date boundaries of the chronological split. **Different:** five weather variables and three demand lags, a 168-hour warm-up, and the definition of eligible samples.

> **Not claimed:** the first weather study, any leakage in prior work, or superiority over probabilistic methods.

Research Aim and Research Questions

> **Aim:** separate the value of **available** weather information from retrospective information, and study model complexity under out-of-time evaluation.

	Research question	Objective	Falsifiable hypothesis	Analyses
RQ1	Does previous-hour observed weather improve next-hour forecasts, and how does target-hour actual weather differ?	Quantify E1 vs. E2 vs. E3	E2 may beat E1; E3 may help precipitation hours but is not guaranteed to beat E2.	X1, X4, X5
RQ2	How do model complexity and regularization affect out-of-time generalization?	Map train vs. validation error	Moderate regularization may help; complex models need not win.	X1, X2, X3
RQ3	What do demand, time, calendar and weather contribute, and where do models fail?	Attribute value and locate errors	Demand history may add most; peak hours may have larger absolute error.	X4, X5

Dataset, Preprocessing and Chronological Split

8,760
hourly observations, 14 columns

8,297
eligible samples

4,152
training

720
validation

3,425
test

Dec 1–7, 2017: 168-h warm-up



1 Align timestamps

Date + Hour become a unique hourly timestamp (Seoul local time); no gaps, no missing cells.

›

2 Build lags first

lag_1, lag_24 and lag_168 are shifted on the full hourly axis; non-functioning hours stay as observed history.

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3 Select eligible rows

Score only Functioning Day = Yes targets with all lags available: 8,465 operating hours, 8,297 after warm-up.

›

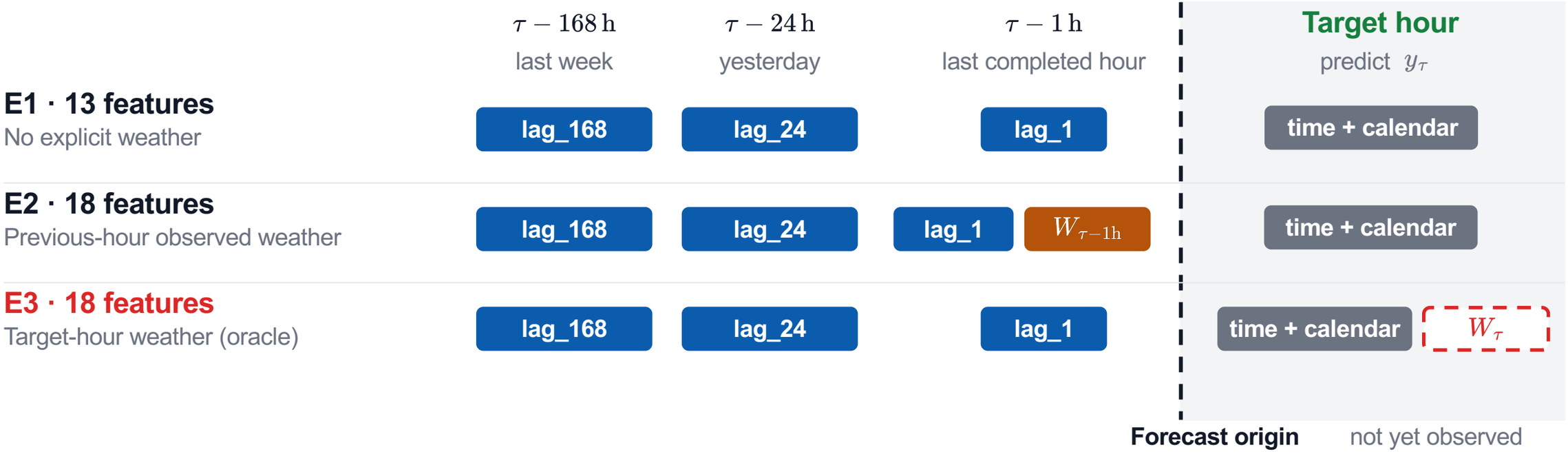
4 Fit on train only

StandardScaler and encoders are fitted on training rows, then applied to validation and test.

Leakage control: chronological split, past-only lags, no target-hour demand input.

Source: Kaggle, Seoul Bike Sharing Demand Prediction (SeoulBikeData.csv); file checksum identical to the UCI release. Bar length proportional to calendar days.

Feature Engineering and Information Settings (E1–E3)



Group	Columns	Encoding	Dim.
H Historical demand	lag_1, lag_24, lag_168	StandardScaler	3
T Time	Hour and Weekday of the target hour	sin / cos pairs	4
C Calendar	Season (4 levels), Holiday (2 levels)	One-hot	6
W Weather	Temperature, Humidity, Wind speed, Rainfall, Snowfall	StandardScaler	5

E1 = H + T + C. E2 adds the previous-hour weather; E3 replaces it with target-hour weather. E3 is a retrospective oracle, is not deployable, and need not beat E2. Target-hour demand is never an input.

Experimental Design: X1–X3

X1–X5 are analyses; E1–E3 are information settings. They are crossed, not eight separate experiments.

Analysis	Models	E1	E2	E3	Varied or controlled
X1 Model comparison	Ridge, KNN, Random Forest, 3 naive	●	●	●	Model × information setting
X2 Ridge regularization	Ridge	○	●	○	Penalty α
X3 Complexity control	KNN, Decision Tree	–	●	–	k; max_depth
X4 Weather and ablation	Ridge, Random Forest	●	●	●	Setting; removed group (E2)
X5 Temporal error analysis	Best E2 model by validation	○	●	○	Hour, month, day type, rain

● main analysis ○ supplementary comparison – not used

X1 · Model comparison

- Does the model ranking change with the information setting?
- Ridge, KNN, Random Forest × E1, E2, E3, plus 3 naive baselines

X2 · Ridge regularization

- How does L2 strength change out-of-time error?
- E2 fixed; 10 log-spaced α; train vs. validation RMSE

X3 · Complexity control

- Does added flexibility only lower training error?
- E2 fixed; KNN k = 1–97; tree max_depth = 2–16, None

X2 Ridge objective

$$\min_{\mathbf{w},b} \sum_{i=1}^{n} \left(y_i - b - \mathbf{x}_i^\top \mathbf{w}\right)^2 + \alpha \|\mathbf{w}\|_2^2$$

Experimental Design: X4–X5

X4 · Weather information and feature ablation

4a E1 vs. E2 vs. E3. Ridge and Random Forest on identical samples.

$$\text{Gain}_{\text{history}} = \text{MAE}(\text{E1}) - \text{MAE}(\text{E2})$$

$$\text{Gap}_{\text{oracle}} = \text{MAE}(\text{E2}) - \text{MAE}(\text{E3})$$

4b Group ablation on E2. Hyperparameters fixed from Full E2; same samples.

Variant	Removed columns
Full E2	none (reference)
− H	lag_1, lag_24, lag_168
− T	hour and weekday sin / cos
− C	season and holiday
− W	5 weather columns (= E1)

$$\Delta\text{MAE}_{\text{remove}}(G) = \text{MAE}(-G) - \text{MAE}(\text{Full E2})$$

X5 · Temporal error analysis

Pre-specified slices of the test period, for the best E2 model by validation MAE:

Slice	Definition
Hour	MAE for each target hour 0–23; night 00–05, peaks 07–09 and 17–19
Month	July to November 2018
Day type	Weekday vs. weekend; holiday flag
Precipitation	Rainfall > 0 or Snowfall > 0 vs. none (scoring label only)
Curve	Actual vs. predicted, 2018-07-01 to 2018-07-07 (fixed in advance)

Each slice reports n, mean rentals, MAE, RMSE and signed error.

> Ablation is not a causal intervention.

Evaluation Protocol, Resources and Project Timeline

Evaluation metrics

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \qquad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

- Unit: rentals per hour; predictions clipped at 0
- No MAPE (zero counts); rolling one-step forecasts

Model selection and leakage control

- Validation MAE selects hyperparameters
- Test evaluated once on a pre-registered list
- Scalers fitted on training rows only; no shuffled K-fold

Project timeline

Oct 26	Oct 27	Week 1	Week 2	Week 3	Week 4	Week 5
Moodle upload by 23:59	Presentation (backup Nov 3)	Shared pipeline, baselines, Ridge	X1–X3 on train / validation; freeze settings	X4 setup; LSTM extension	Freeze list; run test and X5 once	Reproducibility check, final report

Resources and budget

Python, pandas, scikit-learn (Pedregosa et al., 2011), matplotlib on laptop CPUs. No additional budget.

Collaboration

A: data and pipeline. B: Ridge, KNN (X2, X3). C: Random Forest (X1, X4). D: X5, integration, LSTM extension.

Ethics and confidentiality

Public aggregated hourly data; no human subjects, personal data or animal experiments. No ethics review is claimed.

Proposed Outcomes and Research Limitations

Planned analyses and deliverables

- Model × information-setting table of MAE and RMSE, with naive baselines
- Quantified Gain and Gap between E1, E2 and E3
- Ridge α curve; KNN k and tree-depth curves
- Group-ablation chart of MAE change (H, T, C, W)
- Hourly, monthly, weekday and precipitation error analysis
- Reproducible, time-ordered, leakage-checked pipeline

Research limitations

- One city and one year; no multi-year or cross-city claim
- Season and time are confounded in the split
- Aggregate rentals are not station demand or unmet demand
- E2 assumes the previous hour is already published
- E3 is oracle information, not a weather forecast
- Point forecasts only; no interval calibration

Figures shown in the final report will come from executed experiments; none are claimed here.

Contribution: a reproducible protocol separating available from retrospective weather information, with complexity and error analysis.

References

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